Peer Review: "From Global Topology to Navigable Trajectory Space: Mapper Analysis of UK Employment Trajectories"

Reviewed for: *Sociological Methods & Research*
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Reviewer Expertise: Topological Data Analysis and Social Sciences Methodology

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EXECUTIVE SUMMARY

This paper presents a well-executed application of the Mapper algorithm to explore within-regime heterogeneity in UK employment trajectories. The work successfully complements a companion persistent homology paper by revealing local geometric structure invisible to global topological summaries. The sensitivity analysis is particularly thorough and yields important methodological insights about algorithm-dependent versus data-dependent findings.

Overall Assessment: ACCEPT with minor revisions

The paper makes a solid methodological contribution to life-course sociology and demonstrates the utility of TDA beyond hypothesis testing. However, there are notable limitations in causal interpretation, embedding robustness, and some statistical claims that require clarification.

MAJOR COMMENTS

1. The Core Contribution: Clear but Somewhat Incremental

Strength: The paper demonstrates that within-regime heterogeneity exists and persists across 24 configurations. The "causal geography" framing is intuitive and the visualization approach is novel for sociology.

Concern: The core finding—that clusters are internally heterogeneous—is not entirely new in life-course research. Halpin (2010) and Studer (2013) are cited, and low silhouette scores (0.215) in OM clusters already suggested this. What Mapper adds is:

- Visual navigability of heterogeneity (valuable but incremental)
- Persistence across configurations (the novel contribution)
- Outcome-coloured graphs (useful but operationally straightforward)

Recommendation: Strengthen the novelty framing by:

- Explicitly stating that you are the first to apply Mapper to life-course trajectories in sociology
- Clarify that while heterogeneity was suspected, the *geometric structure* of that heterogeneity (the 4-unit PC1 spread within R6) is new
- Add a sentence on why sociologists should care about navigable local structure vs. regime-level summaries (efficiency targeting? theoretical understanding of absorbing states?)

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2. Algorithm Dependence: Important Caveat or Fundamental Problem?

Strength: The sensitivity analysis comparing DBSCAN (50.7% coverage, 0 bridge nodes) to agglomerative clustering (100% coverage, 203–449 bridge nodes) is exemplary and methodologically important.

Critical Issue: The paper frames this as "algorithm properties, not data properties," but the interpretation hinges on which algorithm is "correct."

Questions:

1. **DBSCAN's noise interpretation:** You argue that DBSCAN's 49% noise rate reveals churning trajectories are "geometrically sparse." But is geometric sparsity within a Mapper cover cell truly uninformative? Cover cells are artificial constructs defined by the lens and cover parameters. A trajectory could be sparse at ε=0.5 but form valid clusters at ε=1.0 simply because the cover grid resolution changed.
    - **Suggestion:** Provide a principled argument for which clustering choice is more appropriate for your research question. Is the goal exploratory (use agglomerative for complete coverage) or conservative (use DBSCAN to identify only tight, well-separated clusters)?
2. **Bridge nodes in agglomerative:** You report 203–449 bridge nodes, suggesting regime boundaries are permeable. But these numbers scale dramatically with distance threshold (1.0 → 44,396 nodes; 2.0 → 33,119 nodes). Are regime boundaries truly porous, or are you detecting artifacts of hierarchical merging at different thresholds?
    - **Suggestion:** Plot the relationship between distance threshold and bridge node count more explicitly. Is there a natural elbow or does the relationship appear arbitrary?
3. **Practical recommendation:** Your suggestion to "use DBSCAN for visualization, agglomerative for boundary analysis" is pragmatic but somewhat unsatisfying. Consider proposing a formal criterion for algorithm selection based on downstream application.

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3. Embedding Dependency and Generalization

Concern: You acknowledge (§5.6) that ARI = 0.33 between PCA-20D and UMAP-16D regime assignments, but then say sub-region robustness to alternative embeddings "remains untested." This is a critical gap.

Why this matters:

- If Mapper sub-regions in R6 (the 4-unit PC1 spread) are specific to PCA-20D, the "causal geography" is interpretation-dependent, not data-dependent.
- The paper relies heavily on PC1 as the "income gradient," but this is an artifact of PCA, not necessarily the most policy-relevant dimension.

Recommendations:

1. **Essential:** Test whether the top-3 or top-5 within-regime heterogeneous regimes (R6, R5, R4) maintain similar sub-region structure under UMAP and other embeddings.
2. **Strongly recommended:** Conduct Mapper analysis using the first two UMAP dimensions as the lens function. If sub-regions are robust, this strengthens claims about data properties vs. embedding artifacts.
3. If you cannot run additional embeddings before resubmission, reframe findings as "robust to Mapper configuration, contingent on PCA-20D embedding" rather than simply "robust."

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4. Validity of Outcome Variables and the "Reconstruction" Procedure

Concern: Table 5 reconstructs 14 outcome variables from raw sequences, including employment rate, churning rate, and final income band. The methods section (§3.4) provides minimal detail.

Missing details:

- How is "churning rate" computed? (State transitions per year appears straightforward, but are consecutive identical states treated as no transition?)
- "Employment improvement score" is mentioned in Table 5 but never computed or displayed. What is it?
- "Escape-to-employment indicator" — binary threshold? Defined at what point in the sequence?
- Are outcomes computed over the *entire* trajectory or a subset? (Some respondents have only 10 years; others 14.)

Concern: Figure 7 shows final income band by node. This is visually striking but potentially circular: if nodes are defined partly by PCA dimensions that correlate with income, and income is derived from the same raw states used to compute n-grams, the outcome coloring may reflect the clustering structure rather than revealing new heterogeneity.

Recommendations:

1. **Essential:** Define each outcome variable precisely in §3.4. Include formulas or pseudocode.
2. **Clarify:** Is there any re-use of information between the embedding construction and outcome computation? If so, acknowledge potential circularity and propose outcome variables constructed from *different* data sources (e.g., education, region, demographics) in a revision.
3. **Display:** Create a table showing the correlation matrix between PC1, L2 norm, and the 14 outcome variables. High correlations would suggest overlapping information.

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5. Causal Geography: Metaphor or Claim?

Concern: You introduce "causal geography" as a framing device (§1.2), but this terminology is problematic.

- **Causal language implies intervention potential:** You write (§5.3): "Nodes near the Employed Mid boundary plausibly contain individuals whose churning reflects temporary disruption... nodes near the Inactive Low boundary contain individuals drifting toward the absorbing inactivity state."
- **But Mapper is descriptive:** The graph cannot distinguish between:
  - Individuals trapped in inactivity (absorbing state)
  - Individuals choosing inactivity (retirement, caregiving)
  - Individuals with health barriers to employment

  The spatial clustering of outcomes *correlates* with regime type but does not establish causation.

Recommendation: Relabel "causal geography" as "outcome geography" or "heterogeneous outcome landscape." Clarify in §5.3

that the Mapper graph identifies *where in trajectory space outcomes differ* but does not establish *why*. The subsequent sentence about different interventions becomes a suggestion for future research rather than a conclusion from this analysis.

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6. Statistical Testing and Null Model

Issue: The paper reports descriptive statistics (node means, z-scores) but no formal significance tests. Some specific concerns:

1. **Sub-regime detection (|z| > 1.0):** Why use ±1.0? This is roughly a 68% confidence interval under normality, but employment trajectories are likely skewed. Have you considered FDR correction across 7 regimes × 2 variables (14 tests)? Non-parametric quantiles instead of z-scores?
2. **PC1 range comparisons (§4.3, Table 4):** R6 spans 4 units on PC1; R3 spans 0.13. Is this statistically significant? What is the sampling distribution of PC1 range under null conditions (e.g., random shuffling of trajectories within regime)?
3. **Figure 7 (final income) interpretation:** Visually striking, but lacks statistical summary. What fraction of R6 nodes are in the "mid-income" sub-region vs. "low-income" sub-region? Are these fractions significantly different from regime-level expectations?

Recommendations:

1. In §3.7, justify the |z| > 1.0 threshold or use multiple thresholds (±0.5, ±1.0, ±1.5) with robustness discussion.
2. Report counts and fractions in text; don't rely on visual color maps alone.
3. Consider adding permutation tests: shuffle regime labels within PCA-20D space and recompute Mapper to generate null distribution of sub-regime node counts.

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7. Comparison with Optimal Matching: Underdeveloped

Current treatment (§2.4, §5.5):

- You note ARI = 0.26 between OM and GMM regimes
- You claim OM "produces flat clusters with no internal structure" while Mapper shows internal topology

Problem: This undersells OM's capability. Dendrogram cutting at different heights *does* reveal hierarchical structure. Why not visualize an OM dendrogram coloured by outcome variables alongside your Mapper graph? This would make the comparison concrete.

Recommendation:

1. Display the OM dendrogram with node labels colored by mean employment rate or PC1, using the same visual scheme as Figures 5–7.
2. Compare the *interpretability* of hierarchical OM structure vs. Mapper graph structure for sociology audiences.
3. Discuss computational cost: Mapper requires tuning cover, lens, and clustering; OM requires choosing merge distance threshold. Which is easier for practitioners?

MINOR COMMENTS

8. Methodological Details

Issue A: Appendix or Main Text?
The 24-configuration grid (Table 6–7) is comprehensive but bulky. Consider moving to appendix and reporting only DBSCAN PCA-2D and best-performing agglomerative variant in main text.

Issue B: Figure 9 Interpretation
You state: "R3 and R5 nodes show churn rates of 0.32–0.45 despite low employment, indicating that these individuals are not stably inactive but cycle."

But "cycling" between unemployment states (U) is different from cycling between employment and inactivity (E-I). The n-gram embedding should capture this, but you don't clarify whether high churning in R3/R5 reflects U-U transitions or E-I oscillations.

Issue C: Sample Characteristics
Mean trajectory length 12.9 years with range 10–14 is quite narrow. How much of the variation is due to respondent cohort (born 1930s–1990s) vs. attrition?

How representative is this sample of the UK labour force? Are low-income workers undersampled (as in many panel surveys)?

Recommendations:

1. Clarify the exact state transitions defining "churning" for R3/R5.
2. Add a sample representativeness paragraph in §3.1 or Table 1.

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9. Literature Gaps

Strength: You cite Rizvi et al. (2017) on cell differentiation and Lum et al. (2013) on racial profiling, making a case for Mapper in sociology.

Gap: Limited engagement with recent topological sociology papers. Since 2020, there have been applications of TDA beyond your scope. More importantly, life-course sociology has seen limited TDA adoption. Have you checked recent work on sequence complexity measures?

Recommendation: Extend §2.3 to include post-2015 life-course methods (complexity, entropy-based clustering) to position Mapper as an alternative to optimal matching.

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10. Reproducibility and Code Availability

Issue: You use KeplerMapper 2.1 but no hyperlink or code repository is mentioned. The n-gram embedding, PCA freezing, and outcome reconstruction are not publicly available.

Recommendation:

- State: "Code and reproducible notebooks are available at [GitHub]"
- Or: "Code available upon request from corresponding author"
- Provide at minimum: R or Python script for outcome reconstruction (§3.4)

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11. Figures: PDF Rendering and Legends

From the PDF:

- Figures 1–10 render clearly with informative colour gradients. Good.
- Figures 5–7 (DBSCAN comparison) are small; adding axis labels would help.
- Figure 6 (agglomerative right panel showing top 1,500 nodes): The truncation is pragmatic but undermines claims about bridge nodes. A note should clarify: "For visibility, only the 1,500 largest nodes are displayed; full graph contains [X] nodes."

Recommendation: Add edge labels in Figures 9 and 10 to distinguish connected components from isolated components.

SPECIFIC TECHNICAL CORRECTIONS NEEDED

1. **Equation in §2.1:** "$f : X \to \mathbb{R}^k$" — specify that X is your point cloud (27,280 × 20) and k is typically 1–2.
2. **Table 3, interpretation:** You state "R6 fragments less (80 nodes)" but then note "low coverage (755 of 2,064)." This is ambiguous. Clarify: "R6 is split into 80 clusters under DBSCAN, covering 755 of its 2,064 members (36.6%); the remaining 1,309 are classified as noise."
3. **§4.4.1:** "Escape-to-employment indicator" is listed in Table 5 header but not computed. Either define it and show results, or remove it from the table.
4. **References:** "Author (2026a)" appears throughout but should be replaced with actual author name before submission. Check that all 40+ references are correctly formatted for *Sociological Methods & Research* style.

MISSING INFORMATION

Essential for Publication:

1. Formal outcome variable definitions (§3.4 or appendix)
2. Embedding robustness check (UMAP or alternative embedding, even if partial)
3. Null permutation tests for sub-regime node counts
4. Code availability statement and reproducibility details

Strongly Recommended:

5. Comparison visualization (OM dendrogram vs. Mapper graph)
6. Clarity on causal language (outcome geography vs. causal geography)
7. Algorithm selection criterion (principled choice between DBSCAN and agglomerative)
8. Sample representativeness analysis

OVERALL STRENGTHS

✓ **Thorough sensitivity analysis** comparing 24 Mapper configurations — sets a high bar for applied TDA papers
✓ **Clear writing** with accessible explanation of Mapper for sociology audiences
✓ **Novel application domain** — first systematic Mapper application to UK life-course data
✓ **Outcome-coloured visualizations** — effective communication of heterogeneous outcomes within regimes
✓ **Honest limitations section** — acknowledges embedding dependency, descriptive nature, and visualization challenges

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OVERALL WEAKNESSES

✗ **Embedding robustness untested** — core findings may be PCA artifacts
✗ **Algorithm dependence underexplored** — unclear which approach is appropriate for sociological questions
✗ **Causal framing without causal methods** — "causal geography"

terminology overreaches
✗ **Outcome circularity not addressed** — outcomes derived from same n-grams used for embedding
✗ **Comparison with OM mostly rhetorical** — lacks quantitative, side-by-side comparison
✗ **No formal significance testing** — relies on visual interpretation and z-score flags (±1.0 arbitrary)

VERDICT

This paper merits publication as a methodological contribution demonstrating how Mapper reveals local heterogeneity invisible to global topological summaries. The sensitivity analysis is the strongest aspect and offers practical guidance for other Mapper applications.

However, the paper **should not be accepted as-is**. The following revisions are non-negotiable:

1. Test embedding robustness (UMAP, at minimum)
2. Formalize outcome variable definitions
3. Clarify algorithm selection rationale
4. Remove or reframe "causal geography" terminology

With these revisions, the paper becomes a solid methodological contribution to life-course sociology and an exemplar for TDA applications in social science.

RECOMMENDED REVISIONS BY PRIORITY

Tier 1 (Essential):

- Define outcome variables precisely with pseudocode or formulas
- Provide clear rationale for DBSCAN vs. agglomerative choice
- Test Mapper robustness under alternative embeddings (UMAP)
- Reframe "causal geography" as "outcome geography"
- Add permutation null tests for sub-regime node counts

Tier 2 (Strongly Recommended):

- Compare OM dendrogram and Mapper visualizations directly
- Clarify sample representativeness and attrition
- Add statistical significance thresholds for $|z| > 1.0$ sub-regime detection
- Provide GitHub/code repository link

Tier 3 (Polish):

- Move Tables 6–7 to appendix
- Add clarifying notes to Figure 6 about truncation to top 1,500 nodes
- Fix "Author (2026a)" to actual author name throughout
- Verify all 40+ references match journal style

CLOSING REMARKS

This paper advances TDA applications in sociology and demonstrates important methodological lessons about algorithm dependence in Mapper. The visualization approach and outcome-coloured graphs are novel in this domain. However, the work is held back by unexamined embedding dependence and imprecise causal framing.

The revisions outlined above are straightforward and will significantly strengthen the manuscript. I recommend **conditional acceptance** pending these changes.

Reviewer Confidence: High (very familiar with both TDA and life-course sociology)
Estimated Time to Resubmit: 4–8 weeks for Tier 1 revisions; 2–4 weeks for data-heavy Tier 2 revisions